Andrew Yang

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PROFESSIONAL SUMMARY

Test Engineer on SpaceX's Development Test team. Owns test campaigns end-to-end, from stand design and DAQ through data analysis, for first-of-a-kind (FOAK) hardware across cryogenic, fluid, thermal, and structural systems. Background in nuclear transient analysis; M.S. Mechanical Engineering (Carnegie Mellon); passed the NCEES FE exam; eligible for security clearance.

WORK EXPERIENCE

SpaceX **McGregor, TX**

Test Engineer, Development Test *Jun 2026 – Present*

- Owned design, build, DAQ, and data analysis for a flight-gating butterfly valve test characterizing failure modes under high differential pressure; challenged the original stand concept and proposed an alternate architecture that cut cost ~96% and saved months of schedule.
- Designed the stand with margin to absorb test-matrix changes through test day; resolved same-day issues (e.g., switched pressurant from GN2 to helium) and delivered on time for a flight milestone.
- Leading integration of a permanent LN2 supply for the flight termination system (FTS) test stand, replacing a costly rental (~$96K/yr); designing automated tank control with protective interlock logic and managing fluid quality, O2 monitoring, and truck-access site upgrades.

Westinghouse Electric Company **Pittsburgh, PA**

Transient Analysis Engineer I *Nov 2025 – May 2026*

- Ran safety-critical transient simulations for advanced nuclear systems; developed behavioral models that improved predictive accuracy for off-normal events by 20% over prior failure-rate data.
- Translated simulation results into hardware requirements with design and safety teams, informing subsystem design.

eVinci Test Engineer I / Intern *Aug 2024 – Nov 2025*

- Led 2 and was a key contributor on 3 additional test stands for FOAK heat pipe and fluid systems spanning high-temperature, non-standard-orientation, pump-in-vacuum, and pneumatic-actuator environments.
- Conducted X-ray root-cause investigation of FOAK heat pipe thermal warping; identified the failure mechanism, drove the design correction, and used ANSYS thermal analysis to redefine the instrumentation layout.
- Authored all test procedures and supervised technician execution on large-scale systems; automated the data pipeline in MATLAB/Python, cutting iteration time 15%.
- Integrated LabVIEW DAQ with pressure transducers, thermocouples, flowmeters, and control valves in multiple test setups.

Wabtec Corporation **Erie, PA**

FLX Validation Intern *Summer 2023*

- Validated fire suppression and pneumatic brake systems on a battery-powered locomotive test stand; stepped into production roles during a plant staffing disruption, keeping builds on schedule while supporting validation testing.

ENGINEERING LEADERSHIP

Carnegie Mellon Racing (Formula SAE) **Oct 2021 – Dec 2024**

Welding Captain; Driver Interface System Lead; Chassis Manufacturing & Validation Engineer; Structures Engineer

- Designed and built a torsional stiffness rig to measure chassis rigidity against FEA predictions; validation data drove structural design decisions.
- Designed a modular jig system that cut weld setup time 50% across all weld assemblies; led TIG welding training in steel, aluminum, and titanium for the full fabrication team.
- Engineered a carbon fiber headrest 50% lighter while maintaining structural compliance; streamlined the layup process, cutting production time 15%.

EDUCATION

Carnegie Mellon University **Pittsburgh, PA**

M.S. Mechanical Engineering (GPA: 4.0/4.0) Dec 2024

B.S. Mechanical Engineering (GPA: 3.82/4.0) May 2024

TECHNICAL SKILLS

Test & Instrumentation: Test stand design, validation testing, LabVIEW DAQ, pressure/temperature/flow instrumentation, cryogenic fluids (LOX, LN2) & high-pressure gases (GN2, GHe), control & interlock logic

Analysis & Software: ANSYS, MATLAB, Python, REFPROP, transient simulation

CAD: Siemens NX, SolidWorks, AutoCAD (P&ID)

Fabrication: TIG welding (steel/Al/Ti), CNC & manual machining, composites

Certifications: Passed NCEES FE exam (Sep 2025); in progress: Employee Possessor (EP), explosives handling